“Normal” Speech errors: Part II (the positional level on down)

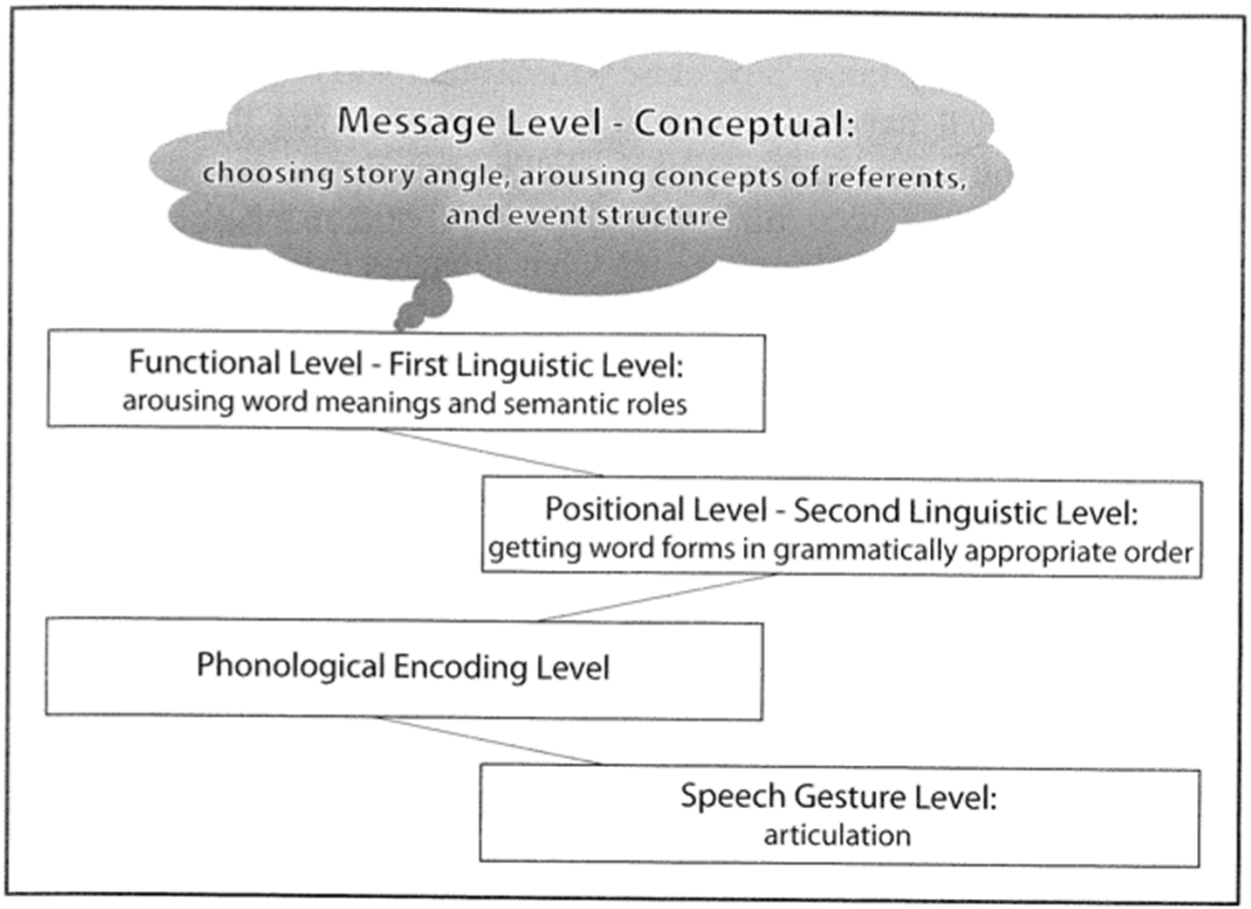


Figure 3.1. Levels of processing discussed in Chapters 3 and 4.

This lecture will cover the bottom three levels of the model:

The Positional Level

The Phonological Encoding Level

The Speech Gestural Level

Then we'll analyze some of the errors you've been sending in and see if we can figure out which level they involve.

An important theme here is that speech errors can provide evidence for the theoretical utility of abstract entities like phonemes, lemmas, noun phrases, etc.

In closing we'll discuss some weaknesses of purely observational research, which will set us up to talk about experimental methods next…

The Positional Level

Let's take the simple-seeming sentence we used as evidence that lemmas are a useful concept:

Where'd I bring it when I put it in?

By the time we're done at the functional level, we know:

1. that we have two Verbs PUT and BRING
2. that I/ME is the Agent of both verbs
3. that IT is the Undergoer of both verbs

What else?
What is the temporal relationship between the verbs?

Expressing all of those things syntactically is what goes on at the *Positional Level.*

If we skip the part where we organize the sentence into a *question*, and work out the appropriate tenses for the verbs, etc., we get something like:

Where'd I [verb, root form] it when I [verb, past] it in?

(Actually, rather a lot of syntactic work had to happen to get that first clause right, because we have both Wh- movement and an auxiliary verb to form the question.)

What we are paying attention to here is that that structure of the first clause requires the base form of the lemma in English.

…and then the next clause requires the past form.

So, what we have at the Positional Level is a syntactic structure with slots for all of the lemmas we activated at the Functional Level, along with all of the other particles required by the grammar.

An error like "Where'd I bring it when I put it in," can be thought of as arising while the slots generated by the Positional Level are being filled with lemmas activated at the Functional level.

A similar error:

a language needer learns

How is this similar to the other sentence?

How is it different?

A similar error:

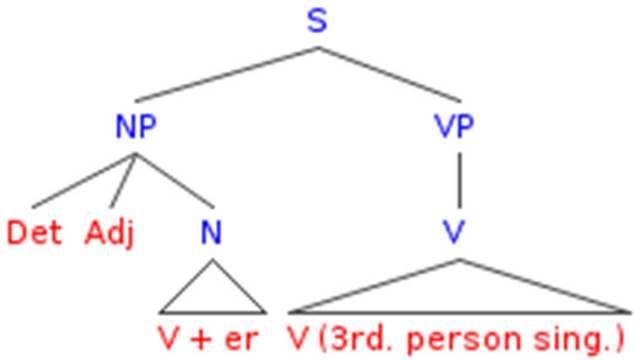
a language needer learns

How is this similar to the other sentence?

two lemmas got switched when they were being fed into the positional level
the positional level stayed basically the same

How is it different?

look at the Noun
it's actually a Verb plus an affix!
*this is **derivational morphology…***



Another similar error:

He did that easy enoughly

How is this similar to “a language needer learns”?

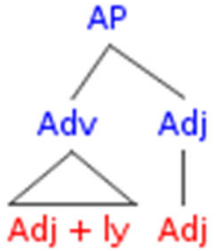
How is it different?

Another similar error:

He did that easy enoughly

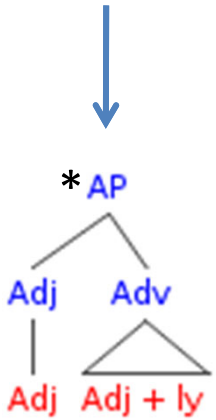
How is this similar to “a language needer learns”?

both have a “stranded affix”
you can make an adjective into an adverb with +ly



How is it different?

in this case, something also happened when setting up the positional level.



This is evidence that derivational morphology is, like*, a thing.*

Specifically *affixes* seem to be able to float around at the Positional Level.

An alternative view might be that derived forms are just other words that are related to their root forms in some rule-like way.

So, you might have a lexicon that looks like this:

needy <> need <> needless

unlearn <> learn <> learner

Etc. That way we wouldn't need to have a separate representation for affixes, and we wouldn't have to remember all of these quirky rules for generating words we use all the time.

But the fact that errors can produce non-words like "needer" and "enoughly" *on the fly*, suggests:

1. Lemmas and affixes are somehow represented separately
2. Affixes have a "slot" in the Positional Level
3. Derived words can be produced by some kind of grammatical process

Consider:

easy → easily /ˈi.zi/ -> /ˈi.zɬl.i/
enough → enoughly /i.ˈnəf/ -> /i.ˈnəf.li/

Remember that derivational morphology can involve complex phonological processes:

certificate /sər.ˈtI.fɬ.kɬt/
certification /ˌsər.tɬ.fɬ.ˈkeI.ʃən/
recertification /ˈri.sər.tɬ.fɬ.ˌkeI.ʃən/

Vowels change, stress moves around, etc. When a process like “add +er” or “make this adjective into an adverb” is specified at the Positional Level, it still leaves some work to do..

Phonological Encoding

This is where we go from a specification of lemmas, affixes, syntactic particles, etc. to a string of phonemes to produce.

So, for “easy enoughly,” the phonological encoding level appears to have worked fine. The speaker did not say “easy /inəvɬli/,” or make some other error in specifying the sequence of phonemes to be produced.

We saw a ton of phonological encoding errors last session, e.g.

fish grotto → gish frotto
Chomsky and Halle → Chomsky and Challe
with this ring I do wed → with this wing I do red

In each case, the problem seems to be specifically with the ordering of phonemes.

Menn describes these errors as “neat.” That is, they follow phonotactic patterns of English, and are (as far as we know) clearly produced.

Speech Gestures

Finally, we go from a sequence of phonemes planned at the phonological encoding level to a series of gestures actually executed by the motor system.

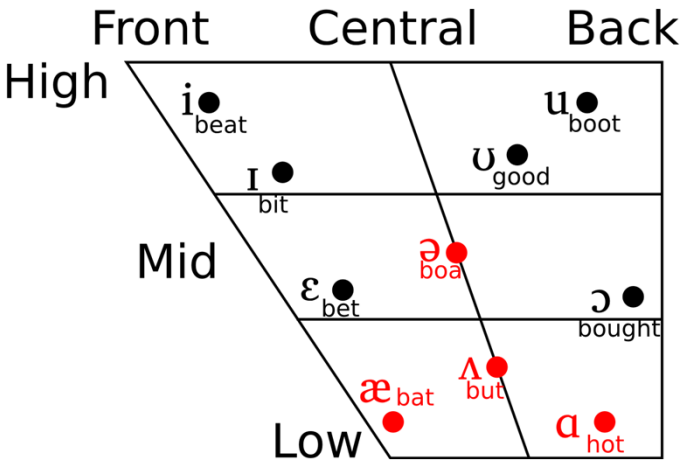
Unlike phonological encoding errors, these are not necessarily “neat.”

Errors at this level result in unclear or imprecise speech.

Note that many errors are ambiguous:

ad hoc → odd hack (/æd hɑk/ -> /ɑd hæk/

This could be metathesis at the Phonological Encoding level, or a miscuing of tongue position at the Gestural Level.



And now for your errors (along with some classics from previous years)…

arising --> "arriving"

it's pretty close to here --> It's pretty [kloʊz] to here

Thoroughly --> "throfly"

buyed in - bought in

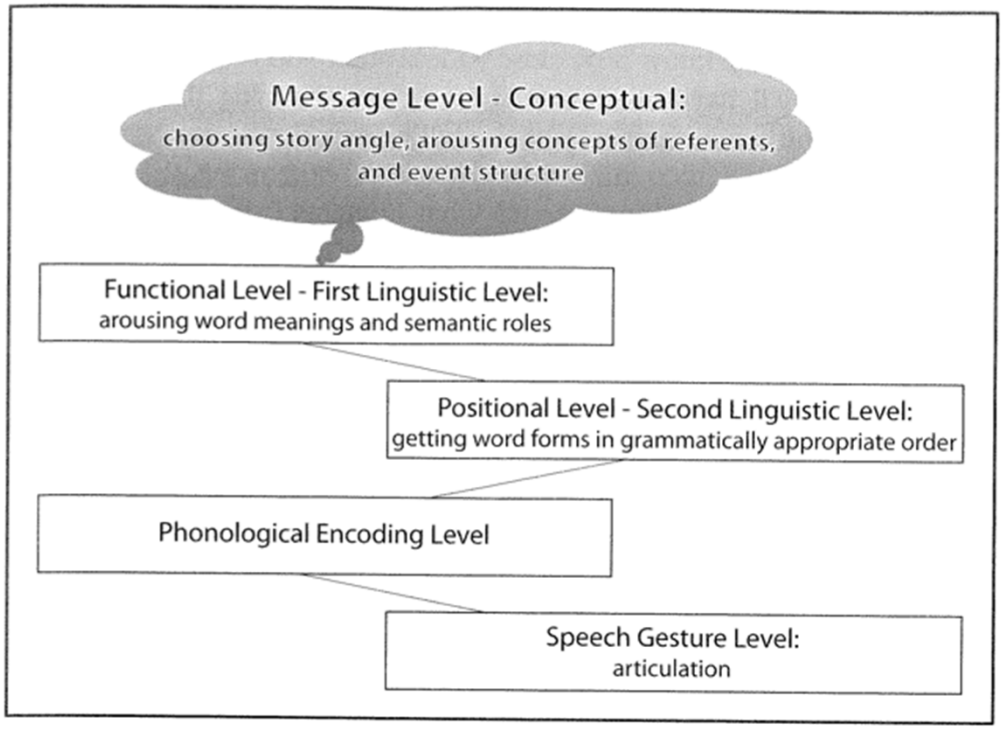


Figure 3.1. Levels of processing discussed in Chapters 3 and 4.

speech-language pathology --> "speech plath... speech plang"

following vowel --> "vollowing fowel"

"Wash my teeth and brush my face."

Lana del Rey --> "Marina del Rey"

"For all intensive purposes."

seat heaters --> "heat seaters"

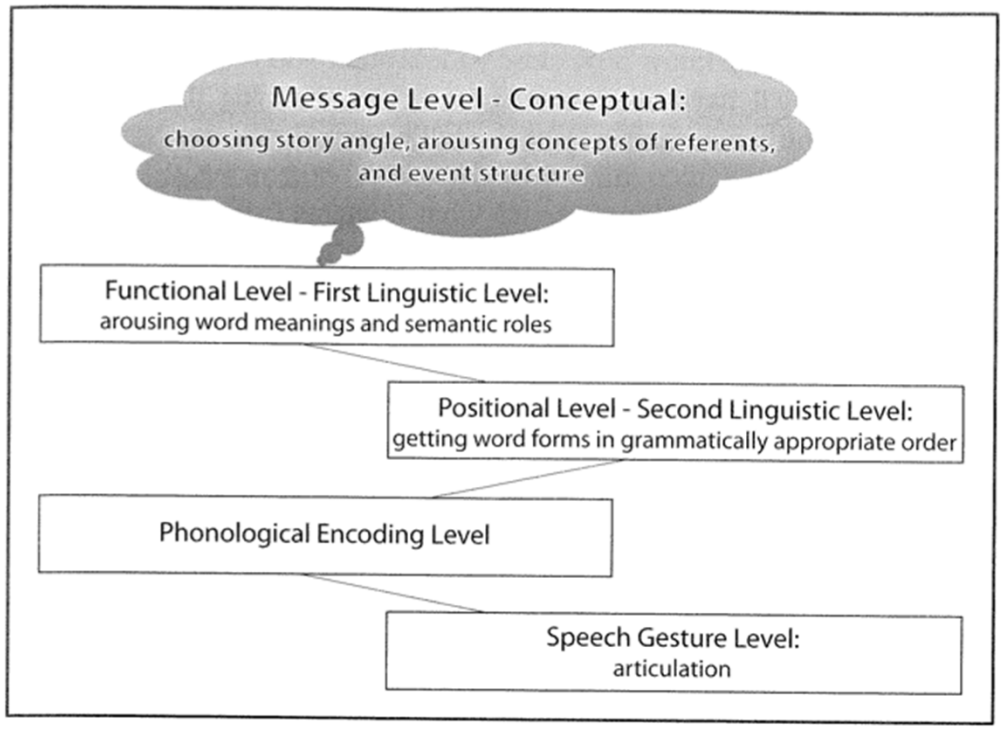


Figure 3.1. Levels of processing discussed in Chapters 3 and 4.

Some greatest hits from previous years:

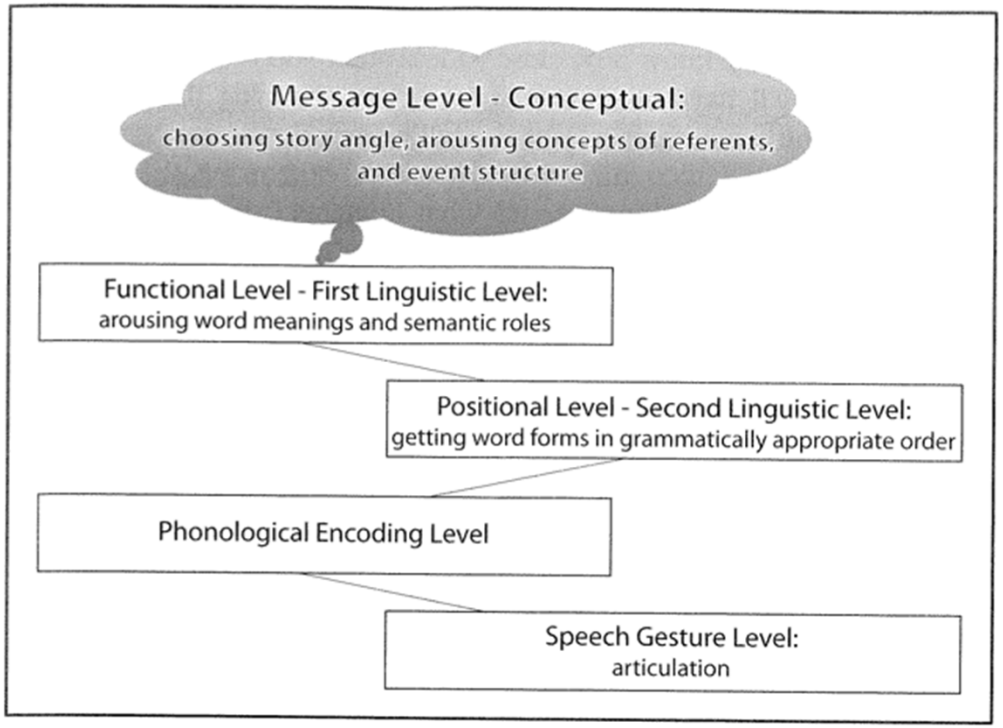


Figure 3.1. Levels of processing discussed in Chapters 3 and 4.

Basically you enter a hundred-dollar room and you do yoga. (re: hot yoga)



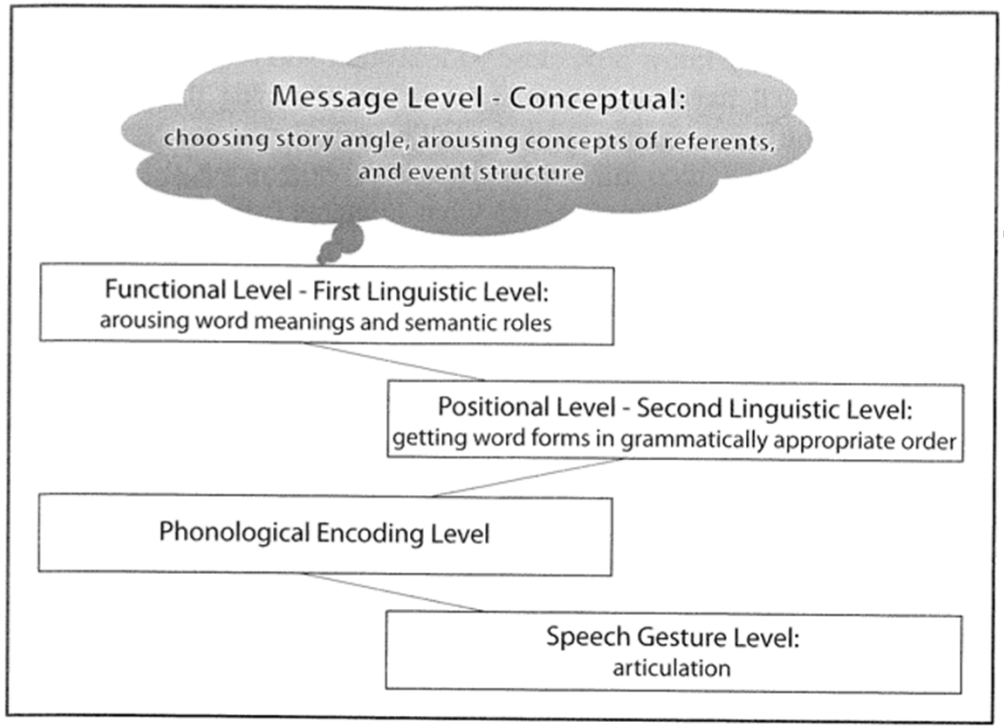
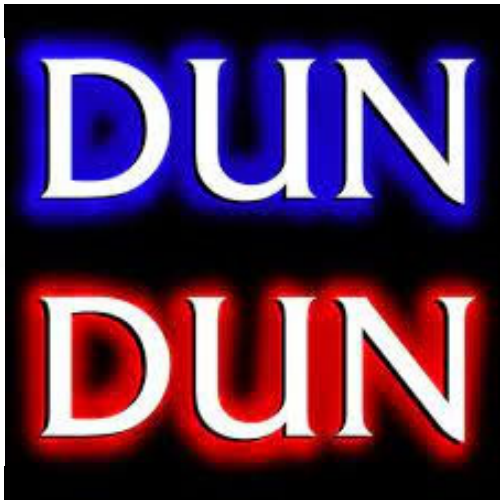


Figure 3.1. Levels of processing discussed in Chapters 3 and 4.

"Law and Order: SUV."





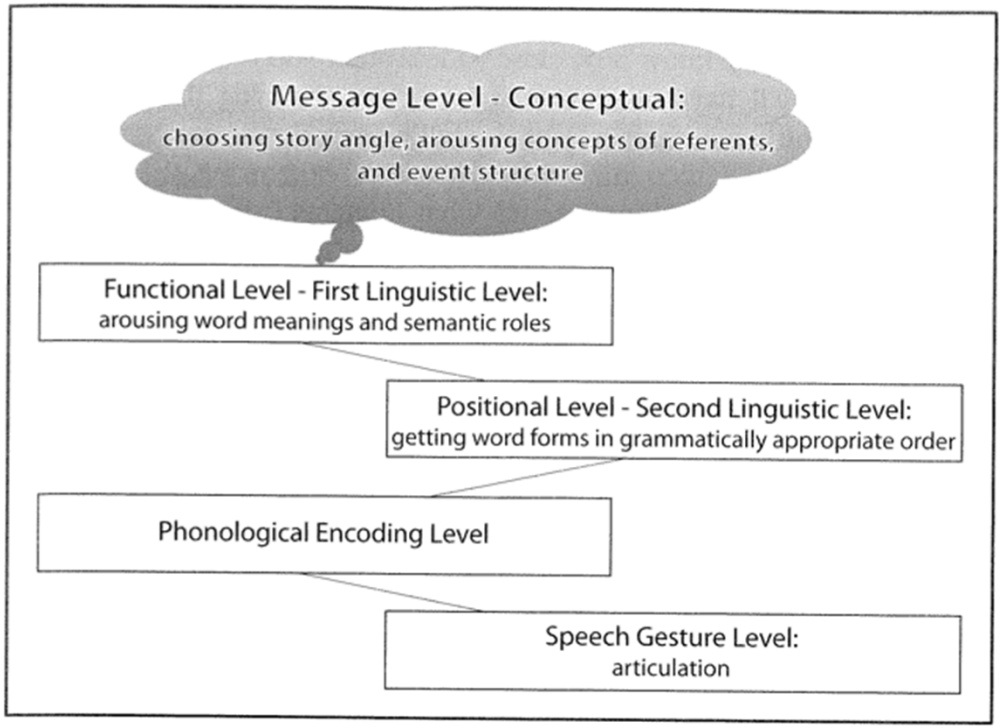


Figure 3.1. Levels of processing discussed in Chapters 3 and 4.

He cooked his kitchen. (chicken)



Some issues with observational studies

Think about how you monitored for speech errors.

What were you listening for?

Did your expectations change as you read the text and learned about speech production in class?

Do you think your expectations might have impacted what you heard? What you reported?

How about when we tried to pinpoint where errors happened in the model just now…

What information about the situation is relevant for reconstructing *how* the error happened?

Some issues with observational studies

In the next few weeks we'll look more closely at speech *perception* and you'll see that:

1. People aren't very good at hearing things that are "halfway" between one phoneme and another, especially for consonants.

2. People are biased against *perceiving* phonotactically illegal strings.

3. People are biased in favor of *perceiving* ambiguous stimuli as words (e.g. something partway between "giss" and "kiss" will *sound* more like "kiss.")

This is what philosophers of science talk about as *theory-dependent observation*. We might miss a lot of errors, or whole classes of errors, because of the way our perceptual systems work.

Experimental methods and careful measurement techniques can remedy *some* of these issues (but not all, not by a long shot!)..

For Wednesday:

Read Menn, Chapter 5 (pp. 153-203)

Extra Stuff

Get ever deeper into speech production. This PNAS paper from Levelt (2001) is a nice summary of how a classic model evolved over the following decades: http://www.pnas.org/content/98/23/13464.full.pdf

A nice review relating production to other areas of research (which, the author notes, isn't done enough! Macdonald, 1999): http://lcnl.wisc.edu/publications/archive/85.pdf